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MATHEMATICAL ENGINEERING DESIGN APPLICATIONS

Gender Bias in Artificial Intelligence Systems for CV Evaluation

FIRST REPORT

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1. INTRODUCTION

Recent studies have increasingly explored the use of artificial intelligence (AI) systems in human resources and recruitment processes. Chen et al. [1] examined AI-based CV screening approaches and noted that automated systems generally achieve faster processing times. Bertrand and Mullainathan [2] demonstrated through field experiments that resume evaluation can be influenced by applicant demographics. O'Connor and Liu [3] analyzed gender bias in AI technologies and identified multiple mechanisms through which such biases can emerge. Navigli et al. [4] investigated biases in large language models, showing that contemporary systems such as ChatGPT contain encodings of social stereotypes. Furthermore, Brookings Institution researchers [5] conducted a large-scale study examining how language models rank resumes and found significant disparities based on applicant demographics.

Research over the past decade has documented gender bias across multiple types of AI applications. Bolukbasi et al. [6] conducted a seminal study demonstrating that word embeddings—the mathematical representations of words underlying most modern natural language processing systems—encode gender stereotypes. In their analysis, they found that word embeddings represented "man" and "computer programmer" as more closely related than "woman" and "computer programmer," despite the lack of any objective basis for this association.

More recently, studies examining contemporary Large Language Models have found evidence of persistent gender bias. Navigli et al. [4] conducted a comprehensive review of biases in large language models and found that current systems exhibit significant gender stereotypes, with substantial variation across different models and different types of tasks. Soundararajan and Delany [7] specifically investigated gender bias in text generation and found that LLMs demonstrate systematic associations between certain genders and particular roles or characteristics.

In the visual domain, Gorska and Jemielniak [8] examined AI-generated images of professionals in STEM fields and documented that image-generating AI systems overwhelmingly depicted STEM professionals as male, thereby reproducing and potentially amplifying gender underrepresentation in these fields.

This body of literature establishes both the theoretical and empirical foundations for the present study, demonstrating that comparative analysis across multiple AI systems represents an important research gap requiring investigation.

1.1. The Importance of the Subject

The integration of artificial intelligence into human resources management has accelerated significantly in recent years. According to McKinsey's 2024 Global AI Survey, over 75% of enterprise organizations now employ some form of AI in their HR processes. Large Language Models (LLMs) such as ChatGPT, Claude, and Google Gemini represent the latest generation of these systems, offering enhanced capabilities for text-based evaluation tasks including CV screening and candidate assessment. The adoption of AI in recruitment offers apparent advantages: increased processing efficiency, reduced time-to-hire, potential cost reduction, and the promise of more objective decision-making. However, alongside this rapid adoption, significant concerns have been raised regarding the fairness and equity of these systems. The most prominent case exemplifying these concerns is Amazon's AI recruiting tool, deployed in 2014 and discontinued in 2018, which systematically discriminated against female applicants. The system, trained on historical hiring data reflecting male dominance in technical roles, learned to associate male applicants with positive outcomes and subsequently penalized female applicants with lower evaluation scores (Dastin, 2018). This case demonstrates that automated evaluation systems, even when developed with good intentions, can perpetuate and amplify existing societal inequalities.

Given that millions of job applicants worldwide are now evaluated at least partially by AI systems, understanding potential gender-based biases in these systems is not merely an academic concern but has direct implications for employment equity, organizational fairness, and regulatory compliance.

1.2. The Purpose of This Study

The primary objective of this study is to measure and compare gender bias in three contemporary Large Language Models—ChatGPT-4, Claude 3 Sonnet, and Google Gemini—when evaluating identical CVs that differ only in the applicant’s name and implied gender. Specifically, the study aims to quantify the magnitude of gender bias in CV evaluation across these state-of-the-art AI systems, determine whether observed differences in evaluation scores between male and female applicants are statistically significant, and identify which of the three systems exhibits the greatest and least gender bias. Furthermore, it seeks to provide empirical evidence that can inform organizational adoption decisions regarding AI-based recruitment tools.

1.3. Scope

This study examines gender bias in CV evaluation using a matched pair methodology within the domain of human resources and recruitment processes. The analysis focuses on three contemporary large language models—ChatGPT-4, Claude 3 Sonnet, and Google Gemini—and evaluates their responses to CV screening tasks for software engineering positions. The primary bias dimension investigated is gender (male/female), using a dataset of 100 CVs constructed as 50 matched pairs that differ only in the applicant's name and implied gender.

In the analysis phase, a matched pair methodology will be employed alongside statistical testing techniques. The evaluation will include descriptive statistics, independent and paired sample t-tests, and effect size analysis to determine both the presence and magnitude of bias across systems.

The outcomes of the study are expected to identify whether significant gender-based differences exist in AI-driven CV evaluation and to determine which system demonstrates relatively higher or lower levels of bias.

1.4 Basic Concepts

To ensure clarity, the following key concepts are defined:

- Artificial Intelligence (AI): Computer systems designed to perform tasks that typically require human intelligence, including learning from data, recognizing patterns, and making decisions.
- Large Language Models (LLM): Advanced AI systems trained on vast amounts of text data to understand and generate human language. Examples include ChatGPT, Claude, and Gemini.
- Gender Bias: Systematic differences in how AI systems treat individuals or groups based on gender, resulting in unequal or inequitable outcomes.
- Matched Pair Methodology: A research design in which pairs of CVs are created that are identical except for one variable of interest (in this case, applicant name and implied gender).
- CV Evaluation Score: A numerical rating (0-100) assigned by an AI system to indicate the suitability of an applicant for a given position.
- Statistical Significance: A determination, typically using p-values, of whether observed differences in data are unlikely to have occurred by chance alone.
- Effect Size: A measure of the magnitude of a difference or relationship, independent of sample size. Cohen's d is used in this study.

1.5. Motivation for Choosing the Topic

This research topic was selected because it focuses on a real-world and high-impact application of artificial intelligence in recruitment processes, allowing for the examination of potential gender bias in widely used large language models such as ChatGPT-4, Claude 3 Sonnet, and Google Gemini, and providing practical insights into the fairness of AI-driven hiring decisions.

2. METHODOLOGY

This study investigates gender bias in AI-based CV evaluation by analyzing score differences across three Large Language Models using a matched pair experimental design.

2.1 Data Pre-Processing

CVs will be selected and standardized to ensure comparability across matched pairs. Each pair will be verified to confirm complete equivalence in content, with the sole variation being the applicant's name and implied gender. Where necessary, CV text will be adjusted to eliminate any unintended gender cues beyond the applicant name.

2.2 Analytical Techniques to be Used

Several statistical and comparative methods will be applied to identify and quantify gender bias:

- Descriptive Statistics will provide an initial overview of score distributions, including means and standard deviations for male and female applicants across each model.

- Independent Samples T-Test will be used to determine whether observed score differences between male and female applicants are statistically significant ($\alpha = 0.05$).

- Cohen's d will measure the practical significance of any detected bias, indicating whether differences are negligible, small, medium, or large in magnitude.

- Cross-Model Comparison will assess whether bias patterns are consistent or vary across ChatGPT-4, Claude 3 Sonnet, and Google Gemini.

2.3 Evaluating Model Bias

The following criteria will be used to assess and compare gender bias across the three systems:

- Bias Magnitude: Mean score difference between male and female applicants per model.
- Statistical Significance: P-value from t-tests indicating whether differences exceed chance.
- Effect Size: Cohen's d to contextualize the practical relevance of observed bias.

These criteria will be applied consistently across all models to enable direct comparison and identify which systems exhibit the most pronounced or significant gender bias.

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